

and Applications
An International Journal

Mittag-Leffler Stability Analysis for a fractional order model of co-contaminated soil and water pollution

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⁴ Lemma 2.1 (Fractional Comparison Principle). [11] Let $u(t) \in C^1([0, T], R_+)$ satisfy: ${}^{ABC}D_t^\alpha u(t) \leq -\lambda u(t)$, $u(0) = u_0$ Then: $u(t) \leq u_0 E_\alpha(-\lambda t^\alpha)$

⁵ Since $0 < \alpha < 1$, we have $\frac{\alpha\pi}{2} < \frac{\pi}{2}$, so $\cos(\frac{\alpha\pi}{2}) > 0$. For real eigenvalues, the stability condition $|\arg(\lambda)| > \frac{\alpha\pi}{2}$ is equivalent to $\lambda < 0$ when λ is real and negative.

⁶ Since all system parameters are positive and at endemic equilibrium $S_p^*, W^*, F^* > 0$, conditions (1) and (4) are automatically satisfied. The remaining conditions depend on the specific parameter values and equilibrium magnitudes.

⁷ Since $C_0 = \frac{aN}{\Phi_C}$ and at equilibrium $aN = \Phi_C C_0$:

⁸ Combining all terms and using LaSalle's invariance principle for fractional systems, we conclude global asymptotic stability when $L \frac{aN}{\Phi_C} < \Phi_2 + \gamma_1$.

⁹ Since at equilibrium $aN = \Phi_C C_0$:

¹⁰ Since $C_0 = \frac{aN}{\Phi_C}$:

¹¹ By the fractional comparison principle:

¹² Since $V(t, X) \geq \frac{1}{2} \|X - E_0\|^2$:

¹³ Fig3: Simulated dynamics of Polluted Soil $S_p(t)$ for $R_0 = 4.55$. The level increases from its initial value and stabilizes at a positive value S_p^* . This confirms the existence and stability of the endemic equilibrium E^* , as $S_p^* > 0$ is a defining characteristic.

¹⁴ Data Availability:

Abstract: Environmental pollution is a serious and long-lasting problem, and understanding how it spreads and how it can be controlled requires robust mathematical models. In this work, we study a five-compartment fractional-order model that describes how pollution moves between different environmental components, namely effluent, soil, water, and farmland. To capture the memory and hereditary effects that naturally occur in environmental processes, we use the Atangana–Baleanu–Caputo (ABC) fractional derivative, which provides a more realistic description than classical integer-order models. We first analyze the qualitative behavior of the model. Using linearization, Lyapunov function, and tools from fractional calculus, we establish the local and global asymptotic stability of two important equilibrium points: the pollution-free equilibrium (E) and the endemic (polluted) equilibrium (E^*). In particular, we apply the concept of Mittag-Leffler stability, which is well-suited to fractional-order systems. A central result of the analysis is the derivation of the basic reproduction number

Keywords: Co-Contamination, ABC mathematical model, fixed point, Adaptive control, Adams-Bashforth.